THE UNIVERSITY OF ZAMBIA

DEPARTMENT OF ECONOMICS

ECN 1215 TUTORIAL SHEET ONE (2016/2017 Second half)

QUESTION ONE

Consider an economy that produces only two goods, bread and Oranges. Below is the data for the respective years.

	2010	**2013**	**2016**
Price of bread	K5	K6.5	K9
Price of oranges/kg	K18	K19	K21
Quantity of bread	1000	1200	900
Quantity of oranges/kg	700	690	500

Using 2010 as your base year, compute the following:

i) Nominal GDP
ii) Real GDP
iii) CPI for each year
iv) GDP Deflator for each year
v) Inflation rate for each year based on your CPI
vi) Describe the trend in inflation.

QUESTION TWO

a) A farmer grows a bushel of wheat and sells it to a miller for $1.00. The miller turns the wheat into flour and then sells the flour to a baker for $3.00. The baker uses the flour to make bread and sells the bread to an engineer for $6.00.The engineer eats the bread. What is the value added by each person? What is GDP?
b) Suppose a man marries his maid. After they are married, his wife continues to perform the home duties as before, and he continues to support her as before (but as a wife rather than as an employee). How does the marriage affect GDP? How should it affect GDP?

QUESTION THREE

One day Yonas Savimbi the plumber collects K400 from his plumbing works. Over this day, his equipment depreciates in value by K50. Of the remaining K350, Yonas Savimbi sends K30 to the government in sales taxes, takes home K220 in wages, and retains K100 in his business to add new equipment in the future. From the k220 that Savimbi takes home, he pays k70 in income taxes. Based on this information, compute Yonas Savimbi's contribution to the following measures of income:

a) gross domestic product
b) net national product
c) national income
d) personal income
e) disposable personal income

QUESTION FOUR

a) With the aid of a well labeled diagram, describe the implication of adding the government activities into the circular flow model.
b) Given a closed economy with no government with the following consumption function

$$C = 100 + 0.75Y$$

Where Y represents National Income and C consumption Expenditure

i) Using this information, complete the table below (S denotes savings)

Y	0	100	200	300	400	600	700	800	900	1000
C										
I	50	50	50	50	50	50	50	50	50	50
S										

ii) Interpret the above consumption function.
iii) From (i) above, compute the marginal propensity to save and interpret your answer.
iv) Using the consumption function above, derive a savings function.
v) Interpret the Investment function in the above table.
vi) Using the data from the above table in (i), draw the savings function, consumption function, investment function and aggregate demand schedules.
vii) From your diagram in part (iii) above, what is the equilibrium level of income?

QUESTION FIVE

a) With a well labeled diagram, explain fully the significance of the 45° line in the Keynesian macroeconomic model, making sure you to explain what the points above and below the 45° line represent.
b) The following information about a Keynesian model of a closed economy without a government is given;
   (i) Autonomous Consumption (A) = 20
   (ii) At equilibrium, consumption spending = 80 and investment spending = 40

   Draw a diagram of the model and clearly indicate the values of the autonomous expenditure and the equilibrium level of income.

c) Yonas Savimbi's disposable income increases from K2,000 to K2,500. As a result, his serving increases from K400 to K 450. Calculate his marginal propensity to consume.

QUESTION SIX

a) Assume a model with expenditure-income identity. If the marginal propensity to save is 0.25, the income tax rate is 0.2 and the marginal propensity to import is 0.15. Calculate the size of the investment multiplier.
b) Calculate the marginal propensity to save for an economy which has a decline in national income of K300 if it had a decline in investment of K150, income tax of 0.25 and it has the marginal propensity to import of 0.17.

c) Given a closed economy without a government sector, calculate the equilibrium level of income of this economy if the level of autonomous consumption is K16 million, the slope of the curve is 0.6 and the level of investment which is also autonomous is K15 million.

QUESTION SEVEN

a) Explain, with the aid of a diagram, the economic significance of the intercept and the slope of the consumption function.
b) Explain, with the aid of a diagram, the three important characteristics of the consumption function.
c) Explain, with the aid of a diagram, the difference between autonomous consumption and induced consumption. Also refer to the meaning of the marginal propensity to consume.
d) Use the diagram or equations to explain the relationship between consumption and saving.

QUESTION EIGHT

a) You are given the following information; autonomous consumption spending = K400 million, investment spending = K100 million, proportional tax rate = $\frac{1}{8}$, marginal propensity to consume = $\frac{6}{7}$, and full employment level of income = 4,800 million. Calculate
   (i) The equilibrium level of income.
   (ii) The budget deficit at the equilibrium level of income.
   (iii) The change in government spending required to attain the full employment income.